3.0 Software Troubleshooting – CompTIA A+ 220-1202

3.1 Troubleshooting Common Windows OS Issues

Issue	Symptoms / Notes	Troubleshooting Steps
Blue Screen of Death (BSOD)	Stop codes, system crashes	Check error code → Update drivers → Run hardware diagnostics → System restore / repair
Degraded Performance	Slow apps, lagging, high CPU/RAM usage	Task Manager → End high-usage processes → Check startup programs → Malware scan → Update OS
Boot Issues	System fails to start or stalls	Check boot order → Repair bootloader → Check BIOS/UEFI → Use recovery media
Frequent Shutdowns	Unexpected power-offs	Check event logs → Test PSU → Check overheating/fans → Update drivers
Services Not Starting	Critical services fail	Check service dependencies → Event Viewer logs → Repair or reinstall service
Applications Crashing	Single or multiple apps	Update/reinstall apps → Check logs → Check compatibility
Low Memory Warnings	RAM usage high	Close apps → Add more RAM → Check for memory leaks

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Issue	Symptoms / Notes	Troubleshooting Steps
USB Controller Resource Warnings	Device not working	Reinstall drivers → Check Device Manager → Test alternate ports
System Instability	Random errors or crashes	Scan for malware → Check drivers → Run SFC / DISM → Event Viewer logs
No OS Found	Bootloader missing / corrupted	Boot repair → Check connections → Reinstall OS if needed
Slow Profile Load	Long login times	Check roaming profiles → Disk cleanup → Antivirus scan
Time Drift	System clock incorrect	Sync time with Internet → Check CMOS battery

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3.2 Troubleshooting Common Mobile OS and Application Issues

Issue	Symptoms / Notes	Troubleshooting Steps
Application Fails to Launch	App crashes immediately	Restart device → Clear cache/data → Reinstall app → Update OS
Application Fails to Close / Crashes	Frozen or unresponsive app	Force stop → Reboot device → Update app → Check storage
Application Fails to Update	Update errors	Check internet → Free space → Clear app store cache → Re-login app store
Application Fails to Install	Errors during install	Check OS version → Free storage → Disable restrictions → Reboot

Issue	Symptoms / Notes	Troubleshooting Steps
Slow to Respond	Lag in UI or apps	Restart → Close background apps → Update OS and apps → Clear cache
OS Fails to Update	Update fails or hangs	Check storage → Check connectivity → Restart → Run update troubleshooter
Battery Life Issues	Rapid drain	Reduce brightness → Disable unused apps → Check background processes → Replace battery if degraded
Random Reboots	Unexpected restarts	Update OS → Check for app conflicts → Run diagnostics → Factory reset if persistent
Connectivity Issues	Bluetooth, Wi-Fi, NFC fail	Toggle Airplane mode → Forget/reconnect network → Update drivers/firmware → Reset network settings
Screen Does Not Autorotate	Orientation locked	Check rotation lock → Calibrate accelerometer → Restart device

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3.3 Troubleshooting Mobile OS and Application Security Issues

Security Concern	Symptoms / Notes	Troubleshooting Steps
Unofficial App Stores	Apps from unknown sources	Uninstall suspicious apps → Enable “Install from Unknown Sources” restrictions
Developer Mode	May expose device	Disable developer mode unless required

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Security Concern	Symptoms / Notes	Troubleshooting Steps
Root / Jailbreak Access	Elevated access risk	Remove root/jailbreak → Restore to factory settings
Unauthorized / Malicious App	App spoofing, malware	Scan with mobile antivirus → Remove suspicious apps → Update OS
High Network Traffic	Slow network / battery drain	Monitor data usage → Identify rogue apps → Limit background data
Degraded Response Time	Apps sluggish	Clear cache → Check for malware → Restart device
Data-Usage Limit Notification	Exceeding limits	Restrict background apps → Adjust sync settings
Limited / No Internet Connectivity	Cannot browse or use apps	Check Wi-Fi/mobile data → Reset network settings → Check router
High Number of Ads	Intrusive ads	Check for adware → Install trusted ad blocker → Remove suspicious apps
Fake Security Warnings	Pop-up alerts claiming infection	Do not click → Scan with antivirus → Clear browser cache
Unexpected App Behavior	App opens, closes, or crashes unexpectedly	Reinstall app → Update OS → Check permissions
Leaked Personal Files / Data	Unauthorized file sharing	Change passwords → Remove malicious apps → Enable device encryption

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3.4 Troubleshooting Common PC Security Issues

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Symptom	Notes / Actions
Unable to Access Network	Check firewall settings → Run network diagnostics → Verify credentials
Desktop Alerts	Could indicate malware or misconfigured security
False Antivirus Alerts	Fake pop-ups or alerts
Altered System or Personal Files	Missing/renamed files
Unwanted Notifications in OS	Pop-ups, ads, alerts
OS Update Failures	Updates cannot complete
Browser-related Symptoms	Random pop-ups, redirects, certificate warnings, degraded performance

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Quick Troubleshooting Checklist

1. **Identify Symptoms** → Determine affected OS or app.
2. **Isolate Cause** → Check logs, safe mode, connectivity.
3. **Resolve Issue** → Update, reinstall, remove malware, or repair OS/app.
4. **Prevent Future Problems** → Enable updates, antivirus, secure configuration, educate users.

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